FLIGHT PASSENGER PREDICTION

VITyarthi AI-ML PROJECT

Airline passenger prediction using Linear Regression and MSE evaluation.

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INTRODUCTION

This project focuses on predicting the number of airline passengers using machine learning techniques. Predicting passenger trends helps airlines plan operations efficiently.

OBJECTIVE

The objective is to build a model that predicts passenger count using year and month data.

METHODOLOGY

The dataset was loaded from an online source. Data preprocessing was done by converting categorical values into numerical form.

A Linear Regression model was used.

The dataset was split into training and testing sets.

DATASET DESCRIPTION

The dataset used in this project contains information about airline passengers over time. It includes features such as year, month, and number of passengers. The dataset was loaded directly from an online source for simplicity and reliability.

MODEL USED: LINEAR REGRESSION

Linear Regression is a supervised machine learning algorithm used to predict continuous values. It works by finding a relationship between input variables (year and month) and the output (passenger count).

CONCEPTS USED

- Mean: Average number of passengers
- Variance: Spread of passenger data
- Mean Squared Error (MSE): Used to measure model accuracy. Mean Squared Error (MSE) is used to evaluate the model. It calculates the average squared difference between predicted and actual values. A lower MSE indicates better model performance.

RESULTS

The model was trained successfully and produced a Mean Squared Error (MSE) of approximately 1415.

The obtained MSE value indicates that the model has a reasonable prediction capability, though there is still some prediction error due to limited features.

CHALLENGES

Initially faced difficulty in handling datasets and understanding code, but resolved by using simple datasets and guided implementation.

FUTURE SCOPE

The model can be improved by adding more features such as holidays, weather conditions, and airline-specific data to increase prediction accuracy

CONCLUSION

This project helped in understanding basic machine learning concepts and applying them to real-world data.

REFERENCES

- Google Colab
- Python libraries (Pandas, Scikit-learn)